Chapter 2

4.

a. The $2^4 = 16$ possible outcomes have been numbered here for later reference.

Outcome	Home Mortgage Number 1	2	3	4
1	*F*	*F*	*F*	*F*
2	*F*	*F*	*F*	*V*
3	*F*	*F*	*V*	*F*
4	*F*	*F*	*V*	*V*
5	*F*	*V*	*F*	*F*
6	*F*	*V*	*F*	*V*
7	*F*	*V*	*V*	*F*
8	*F*	*V*	*V*	*V*
9	*V*	*F*	*F*	*F*
10	*V*	*F*	*F*	*V*
11	*V*	*F*	*V*	*F*
12	*V*	*F*	*V*	*V*
13	*V*	*V*	*F*	*F*
14	*V*	*V*	*F*	*V*
15	*V*	*V*	*V*	*F*
16	*V*	*V*	*V*	*V*

b. Outcome numbers 2, 3, 5, 9 above.

c. Outcome numbers 1, 16 above.

d. Outcome numbers 1, 2, 3, 5, 9 above.

e. In words, the union of (c) and (d) is the event that either all of the mortgages are variable, or that at most one of them is variable-rate: outcomes 1, 2, 3, 5, 9, 16. The intersection of (c) and (d) is the event that all of the mortgages are fixed-rate: outcome 1.

f. The union of (b) and (c) is the event that either exactly three are fixed, or that all four are the same: outcomes 1, 2, 3, 5, 9, 16. The intersection of (b) and (c) is the event that exactly three are fixed and all four are the same type. This cannot happen (the events have no outcomes in common), so the intersection of (b) and (c) is $\varnothing$.

6.

a. $S = \{123, 124, 125, 213, 214, 215, 13, 14, 15, 23, 24, 25, 3, 4, 5\}$.

b. $A = \{3, 4, 5\}$.

c. $B = \{125, 215, 15, 25, 5\}$.

d. $C = \{23, 24, 25, 3, 4, 5\}$.

12.

a. No, this is not possible. Since event $A \cap B$ is contained within event B, it must be the case that $P(A \cap B) \leq P(B)$. However, $.5 > .4$.

b. By the addition rule, $P(A \cup B) = \cancel{.5}\ .6 + .4 - .3 = \cancel{.6}\ .7$.

c. $P(\text{neither } A \text{ nor } B) = P(A' \cap B') = P((A \cup B)') = 1 - P(A \cup B) = 1 - \cancel{.6}\ .7 = \cancel{.4}\ .3$.

d. The event of interest is $A \cap B'$; from a Venn diagram, we see $P(A \cap B') = P(A) - P(A \cap B) = \cancel{.3}\ .6 - .3 = \cancel{.1}\ .3$

e. From a Venn diagram, we see that the probability of interest is $P(\text{exactly one}) = P(\text{at least one}) - P(\text{both}) = P(A \cup B) - P(A \cap B) = \cancel{.6}\ .7 - .3 = \cancel{.3}\ .4$

14. Let A = an adult consumes coffee and B = an adult consumes carbonated soda. We're told that $P(A) = .55$, $P(B) = .45$, and $P(A \cup B) = .70$.

a. The addition rule says $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, so $.70 = .55 + .45 - P(A \cap B)$ or $P(A \cap B) = .55 + .45 - .70 = .30$.

b. There are two ways to read this question. We can read "does not (consume at least one)," which means the adult consumes neither beverage. The probability is then $P(\text{neither } A \text{ nor } B) = P(A' \cap B') = 1 - P(A \cup B) = 1 - .70 = .30$.

The other reading, and this is presumably the intent, is "there is at least one beverage the adult does not consume, i.e. $A' \cup B'$. The probability is $P(A' \cup B') = 1 - P(A \cap B) = 1 - .30$ from **a** = .70. (It's just a coincidence this equals $P(A \cup B)$.)

Both of these approaches use *deMorgan's laws*, which say that $P(A' \cap B') = 1 - P(A \cup B)$ and $P(A' \cup B') = 1 - P(A \cap B)$.

18. The only reason we'd need at least two selections to find a \$10 bill is if the <u>first</u> selection was <u>not</u> a \$10 bill bulb. There are 4 + 6 = 10 non-\$10 bills out of 5 + 4 + 6 = 15 bills in the wallet, so the probability of this event is simply 10/15, or 2/3.

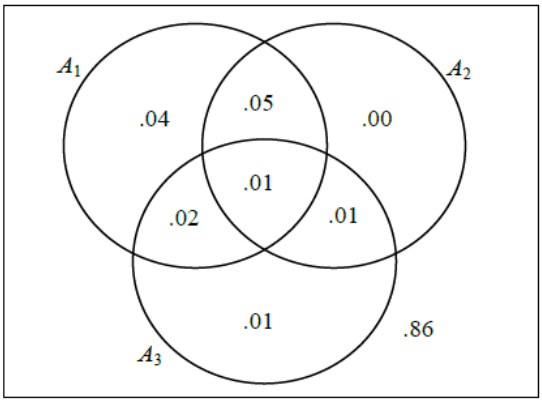
26. These questions can be solved algebraically, or with the Venn diagram below.

a. $P(A_1') = 1 - P(A_1) = 1 - .12 = .88$.

b. The addition rule says $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Solving for the intersection ("and") probability, you get $P(A_1 \cap A_2) = P(A_1) + P(A_2) - P(A_1 \cup A_2) = .12 + .07 - .13 = .06$.

c. A Venn diagram shows that $P(A \cap B') = P(A) - P(A \cap B)$. Applying that here with $A = A_1 \cap A_2$ and $B = A_3$, you get $P([A_1 \cap A_2] \cap A_3') = P(A_1 \cap A_2) - P(A_1 \cap A_2 \cap A_3) = .06 - .01 = .05$.

d. The event "at most two defects" is the complement of "all three defects," so the answer is just $1 - P(A_1 \cap A_2 \cap A_3) = 1 - .01 = .99$.



30.

a. Because order is important, we'll use $P_{3,8} = (8)(7)(6) = 336$.

b. Order doesn't matter here, so we use $\binom{30}{6} = 593{,}775$.

c. The number of ways to choose 2 zinfandels from the 8 available is $\binom{8}{2}$. Similarly, the number of ways to choose the merlots and cabernets are $\binom{10}{2}$ and $\binom{12}{2}$, respectively. Hence, the total number of options (using the Fundamental Counting Principle) equals $\binom{8}{2}\binom{10}{2}\binom{12}{2} = (28)(45)(66) = 83{,}160$.

d. The numerator comes from part **c** and the denominator from part **b**: $\frac{83{,}160}{593{,}775} = .140$.

e. We use the same denominator as in part **d**. The number of ways to choose all zinfandel is $\binom{8}{6}$, with similar answers for all merlot and all cabernet. Since these are disjoint events, $P(\text{all same}) = P(\text{all zin}) + P(\text{all merlot}) + P(\text{all cab}) = \dfrac{\binom{8}{6} + \binom{10}{6} + \binom{12}{6}}{\binom{30}{6}} = \dfrac{1162}{593{,}775} = .002$.

34.

a. Since order doesn't matter, the number of ways to randomly select 5 keyboards from the 25 available is $\binom{25}{5} = 53{,}130$.

b. Sample in two stages. First, there are 6 keyboards with an electrical defect, so the number of ways to select exactly 2 of them is $\binom{6}{2}$. Next, the remaining 5 – 2 = 3 keyboards in the sample must have mechanical defects; as there are 19 such keyboards, the number of ways to randomly select 3 is $\binom{19}{3}$. So, the number of ways to achieve both of these in the sample of 5 is the product of these two counting numbers: $\binom{6}{2}\binom{19}{3} = (15)(969) = 14{,}535$.

c. Following the analogy from **b**, the number of samples with exactly 4 mechanical defects is $\binom{19}{4}\binom{6}{1}$, and the number with exactly 5 mechanical defects is $\binom{19}{5}\binom{6}{0}$. So, the number of samples with <u>at least</u> 4 mechanical defects is $\binom{19}{4}\binom{6}{1} + \binom{19}{5}\binom{6}{0}$, and the probability of this event is

$$\frac{\binom{19}{4}\binom{6}{1}+\binom{19}{5}\binom{6}{0}}{\binom{25}{5}} = \frac{34{,}884}{53{,}130} = .657.$$ (The denominator comes from **a**.)

36. There are $\binom{5}{2} = 10$ possible ways to select the positions for *B*'s votes: *BBAAA, BABAA, BAABA, BAAAB, ABBAA, ABABA, ABAAB, AABBA, AABAB*, and *AAABB*. Only the last two have *A* ahead of *B* throughout the vote count. Since the outcomes are equally likely, the desired probability is 2/10 = .20.

50.

a. $P(\mathbf{M} \cap \mathbf{LS} \cap \mathbf{PR}) = .05$, directly from the table of probabilities.

b. $P(\mathbf{M} \cap \mathbf{Pr}) = P(\mathbf{M} \cap \mathbf{LS} \cap \mathbf{PR}) + P(\mathbf{M} \cap \mathbf{SS} \cap \mathbf{PR}) = .05 + .07 = .12$.

c. $P(\mathbf{SS})$ = sum of 9 probabilities in the **SS** table = .56. $P(\mathbf{LS}) = 1 - .56 = .44$.

d. From the two tables, $P(\mathbf{M}) = .08 + .07 + .12 + .10 + .05 + .07 = .49$. $P(\mathbf{Pr}) = .02 + .07 + .07 + .02 + .05 + .02 = .25$.

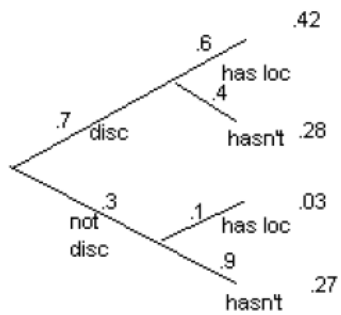
e. $P(\mathbf{M}|\mathbf{SS} \cap \mathbf{Pl}) = \dfrac{P(\mathbf{M} \cap \mathbf{SS} \cap \mathbf{Pl})}{P(\mathbf{SS} \cap \mathbf{Pl})} = \dfrac{.08}{.04+.08+.03} = .533$.

f. $P(\mathbf{SS}|\mathbf{M} \cap \mathbf{Pl}) = \dfrac{P(\mathbf{SS} \cap \mathbf{M} \cap \mathbf{Pl})}{P(\mathbf{M} \cap \mathbf{Pl})} = \dfrac{.08}{.08+.10} = .444$. $P(\mathbf{LS}|\mathbf{M} \cap \mathbf{Pl}) = 1 - P(\mathbf{SS}|\mathbf{M} \cap \mathbf{Pl}) = 1 - .444 = .556$.

56. $$P(A \mid B) + P(A' \mid B) = \frac{P(A \cap B)}{P(B)} + \frac{P(A' \cap B)}{P(B)} = \frac{P(A \cap B) + P(A' \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

58. $$P(A \cup B \mid C) = \frac{P[(A \cup B) \cap C)}{P(C)} = \frac{P[(A \cap C) \cup (B \cap C)]}{P(C)} = \frac{P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)}{P(C)} = P(A \mid C) + P(B \mid C) - P(A \cap B \mid C)$$

60. The tree diagram below shows the probability for the four disjoint options; e.g., P(the flight is discovered and has a locator) = P(discovered)P(locator | discovered) = (.7)(.6) = .42.



a. $P(\text{not discovered} \mid \text{has locator}) = \dfrac{P(\text{not discovered} \cap \text{has locator})}{P(\text{has locator})} = \dfrac{.03}{.03 + .42} = .067$.

b. $P(\text{discovered} \mid \text{no locator}) = \dfrac{P(\text{discovered} \cap \text{no locator})}{P(\text{no locator})} = \dfrac{.28}{.55} = .509$.

74. Using subscripts to differentiate between the selected individuals,
$P(O_1 \cap O_2) = P(O_1)P(O_2) = (.45)(.45) = .2025$.
P(two individuals match) = $P(A_1 \cap A_2) + P(B_1 \cap B_2) + P(AB_1 \cap AB_2) + P(O_1 \cap O_2) = .40^2 + .11^2 + .04^2 + .45^2 = .3762$.

78. P(at least one opens) = 1 – P(none open) = $1 - (.04)^5 = .999999897$.
P(at least one fails to open) = 1 – P(all open) = $1 - (.96)^5 = .1846$.

Chapter 3

non-0's

4. Since a 4-digit code can have between zero and four ~~0s~~, the possible values of X are 0, 1, 2, 3, 4. As examples, the PIN 9876 yields $X = \cancel{0}$ 4 ~~(no 0s)~~, 1006 corresponds to $X = 2$ (two ~~0s~~ non-zeros), and the very poor PIN choice of 0000 implies that $X = \cancel{4}$ 0.

6. The possible X values are1, 2, 3, 4, … (all positive integers). Some examples are:

Outcome:	*RL*	*AL*	*RAARL*	*RRRRL*	*AARRL*
X:	2	2	5	5	5

10.

a. Possible values of T are: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.

b. Possible values of X are: –4, –3, –2, –1, 0, 1, 2, 3, 4, 5, 6.

c. Possible values of U are: 0, 1, 2, 3, 4, 5, 6.

d. Possible values of Z are: 0, 1, 2.

12.

a. Since there are 50 seats, the flight will accommodate all ticketed passengers who show up as long as there are no more than 50. $P(Y \leq 50) = .05 + .10 + .12 + .14 + .25 + .17 = .83$.

b. This is the complement of part **a**: $P(Y > 50) = 1 - P(Y \leq 50) = 1 - .83 = .17$.

c. If you're the first standby passenger, you need no more than 49 people to show up (so that there's space left for you). $P(Y \leq 49) = .05 + .10 + .12 + .14 + .25 = .66$. On the other hand, if you're <u>third</u> on the standby list, you need no more than 47 people to show up (so that, even with the two standby passengers ahead of you, there's still room). $P(Y \leq 47) = .05 + .10 + .12 = .27$.

14.

a. As the hint indicates, the sum of the probabilities must equal 1. Applied here, we get $\sum_{y=1}^{5} p(y) = k[1+2+3+4+5] = 15k = 1 \Rightarrow k = \frac{1}{15}$. In other words, the probabilities of the five y-values are $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{4}{15}, \frac{5}{15}$.

b. $P(Y \le 3) = P(Y = 1, 2, 3) = \frac{1}{15} + \frac{2}{15} + \frac{3}{15} = \frac{6}{15} = .4$.

c. $P(2 \le Y \le 4) = P(Y = 2, 3, 4) = \frac{2}{15} + \frac{3}{15} + \frac{4}{15} = \frac{9}{15} = .6$.

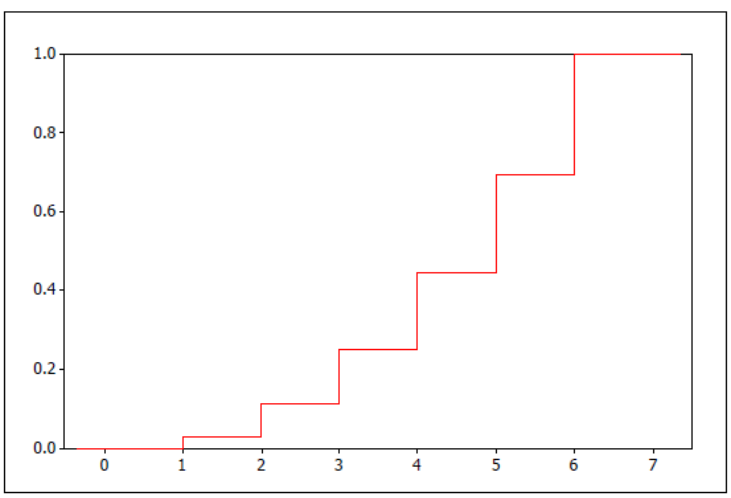
d. Do the probabilities total 1? Let's check: $\sum_{y=1}^{5}\left(\frac{y^2}{50}\right) = \frac{1}{50}[1+4+9+16+25] = \frac{55}{50} \ne 1$. No, that formula cannot be a pmf.

18.

a. $p(1) = P(M = 1) = P(\{(1,1)\}) = \frac{1}{36}$; $p(2) = P(M = 2) = P(\{(1,2)(2,1)(2,2)\}) = \frac{3}{36}$; $p(3) = P(M = 3) = P(\{(1,3)(2,3)(3,1)(3,2)(3,3)\}) = \frac{5}{36}$. Continuing the pattern, $p(4) = \frac{7}{36}$, $p(5) = \frac{9}{36}$, and $p(6) = \frac{11}{36}$.

b. Using the values in **a**,

$$F(m) = \begin{cases} 0 & m < 1 \\ \frac{1}{36} & 1 \le m < 2 \\ \frac{4}{36} & 2 \le m < 3 \\ \frac{9}{36} & 3 \le m < 4 \\ \frac{16}{36} & 4 \le m < 5 \\ \frac{25}{36} & 5 \le m < 6 \\ 1 & m \ge 6 \end{cases}$$



28. If $x_1 < x_2$, then $F(x_2) = P(X \le x_2) = P(\{X \le x_1\} \cup \{x_1 < X \le x_2\}) = P(X \le x_1) + P(x_1 < X \le x_2)$. Since all probabilities are non-negative, this is $\ge P(X \le x_1) + 0 = P(X \le x_1) = F(x_1)$. That is, $x_1 < x_2$ implies $F(x_1) \le F(x_2)$, QED.

Looking at the proof above, $F(x_1) = F(x_2)$ iff $P(x_1 < X \le x_2) = 0$.

32.

a. $E(X) = (16)(.2) + (18)(.5) + (20)(.3) = 18.2$ ft^3; $E(X^2) = (16)^2(.2) + (18)^2(.5) + (20)^2(.3) = 333.2$. Put these together, and $V(X) = E(X^2) - [E(X)]^2 = 333.2 - (18.2)^2 = 1.96$.

b. Use the linearity/rescaling property: $E(70X - 650) = 70\mu - 650 = 70(18.2) - 650 = \$624$. Alternatively, you can figure out the price for each of the three freezer types and take the weighted average.

c. Use the linearity/rescaling property again: $V(70X - 650) = 70^2\sigma^2 = 70^2(1.96) = 9604$. (The 70 gets squared because variance is itself a square quantity.)

d. We cannot use the rescaling properties for $E(X - .008X^2)$, since this isn't a linear function of X. However, since we've already found both $E(X)$ and $E(X^2)$, we may as well use them: the expected actual capacity of a freezer is $E(X - .008X^2) = E(X) - .008E(X^2) = 18.2 - .008(333.2) = 15.5344$ ft^3. Alternatively, you can figure out the actual capacity for each of the three freezer types and take the weighted average.

36. You have to be careful here: if \$0 damage is incurred, then there's no deductible for the insured driver to pay! Here's one approach: let $h(X)$ = the amount paid by the insurance company on an accident claim, which is \$0 for a "no damage" event and \$500 less than actual damages ($X - 500$) otherwise. The pmf of $h(X)$ looks like this:

x	0	1000	5000	10000
$h(x)$	0	500	4500	9500
$p(x)$	.8	.1	.08	.02

Based on the pmf, the average payout across these types of accidents is $E(h(X)) = 0(.8) + 500(.1) + 4500(.08) + 9500(.02) = \$600$. If the insurance company charged \$600 per client, they'd break even (a bad idea!). To have an expected profit of \$100 — that is, to have a *mean* profit of \$100 per client — they should charge \$600 + \$100 = \$700.

42.

a. $E[X(X-1)] = E(X^2) - E(X) \Rightarrow E(X^2) = E[X(X-1)] + E(X) = 27.5 + 5 = 32.5$.

b. $V(X) = E(X^2) - [E(X)]^2 = 32.5 - (5)^2 = 7.5$.

c. Substituting **a** into **b**, $V(X) = E[X(X-1)] + E(X) - [E(X)]^2$.

48. $X \sim \text{Bin}(25, .05)$

a. $P(X \le 3) = B(3;25,.05) = .966$, while $P(X < 3) = P(X \le 2) = B(2;25,.05) = .873$.

b. $P(X \ge 4) = 1 - P(X \le 3) = 1 - B(3;25,.05) = .1 - .966 = .034$.

c. $P(1 \le X \le 3) = P(X \le 3) - P(X \le 0) = .966 - .277 = .689$.

d. $E(X) = np = (25)(.05) = 1.25$, $\sigma_X = \sqrt{np(1-p)} = \sqrt{25(.05)(.95)} = 1.09$.

e. With $n = 50$, $P(X = 0) = \binom{50}{0}(.05)^0(.95)^{50} = (.95)^{50} = .077$.

50. Let X be the number of faxes, so $X \sim \text{Bin}(25, .25)$.

a. $P(X \le 6) = B(6;25,.25) = .561$.

b. $P(X = 6) = b(6;25,.25) = .183$.

c. $P(X \ge 6) = 1 - P(X \le 5) = 1 - B(5;25,.25) = .622$.

d. $P(X > 6) = 1 - P(X \le 6) = 1 - .561 = .439$.

52. Let X be the number of students who want a new copy, so $X \sim \text{Bin}(n = 25, p = .3)$.

a. $E(X) = np = 25(.3) = 7.5$ and $SD(X) = \sqrt{np(1-p)} = \sqrt{25(.3)(.7)} = 2.29$.

b. Two standard deviations from the mean converts to $7.5 \pm 2(2.29) = 2.92$ & 12.08. For X to be <u>more</u> than two standard deviations from the means requires $X < 2.92$ or $X > 12.08$. Since X must be a non-negative integer, $P(X < 2.92 \text{ or } X > 12.08) = 1 - P(2.92 \le X \le 12.08) = 1 - P(3 \le X \le 12) =$

$$1 - \sum_{x=3}^{12} \binom{25}{x} (.3)^x (.7)^{25-x} = 1 - .9736 = .0264.$$

c. If $X > 15$, then more people want new copies than the bookstore carries. At the other end, though, there are $25 - X$ students wanting used copies; if $25 - X > 15$, then there aren't enough used copies to meet demand.

The inequality $25 - X > 15$ is the same as $X < 10$, so the bookstore can't meet demand if either $X > 15$ or $X < 10$. All 25 students get the type they want iff $10 \le X \le 15$:

$$P(10 \le X \le 15) = \sum_{x=10}^{15} \binom{25}{x} (.3)^x (.7)^{25-x} = .1890.$$

d. The bookstore sells X new books and $25 - X$ used books, so total revenue from these 25 sales is given by $h(X) = 100(X) + 70(25 - X) = 30X + 1750$. Using linearity/rescaling properties, expected revenue equals $E(h(X)) = E(30X + 1750) = 30\mu + 1750 = 30(7.5) + 1750 = \$1975$.

54. Let X equal the number of customers who choose an oversize racket, so $X \sim \text{Bin}(10, .60)$.

a. $P(X \ge 6) = 1 - P(X \le 5) = 1 - B(5;20,.60) = 1 - .367 = .633$.

b. $\mu = np = 10(.6) = 6$ and $\sigma = \sqrt{10(.6)(.4)} = 1.55$, so $\mu \pm \sigma = (4.45, 7.55)$.
$P(4.45 < X < 7.55) = P(5 \le X \le 7) = P(X \le 7) - P(X \le 4) = .833 - .166 = .667$.

c. This occurs iff between 3 and 7 customers want the oversize racket (otherwise, one type will run out early). $P(3 \le X \le 7) = P(X \le 7) - P(X \le 2) = .833 - .012 = .821$.

56. Let X = the number of students in the sample needing accommodation, so $X \sim \text{Bin}(25, .02)$.

a. $P(X = 1) = 25(.02)(.98)^{24} = .308$.

b. $P(X \ge 1) = 1 - P(X=0) = 1 - (.98)^{25} = 1 - .603 = .397$.

c. $P(X \ge 2) = 1 - P(X \le 1) = 1 - [.603 + .308] = .089$.

d. $\mu = 25(.02) = .5$ and $\sigma = \sqrt{25(.02)(.98)} = \sqrt{.49} = .7$, so $\mu \pm \sigma = (-.9, 1.9)$. $P(-.9 \le X \le 1.9) =$ $P(X = 0 \text{ or } 1) = .911$.

e. $\dfrac{.5(4.5) + 24.5(3)}{25} = 3.03$ hours. Notice the sample size of 25 is actually irrelevant.

60. Using the hint, $h(X) = 1 \cdot X + 2.25(25 - X) = 62.5 - 1.5X$, which is a linear function. Since the mean of X is $E(X) = np = (25)(.6) = 15$, $E(h(X)) = 62.5 - 1.5E(X) = 62.5 - 1.5(15) = \$40$.

66.

a. Let Y = the number with reservations who show up, a binomial rv with $n = 6$ and $p = .8$. Since there are only four spaces available, at least one individual cannot be accommodated if Y is more than 4. The desired probability is $P(Y = 5 \text{ or } 6) = b(5; 6, .8) + b(6;6,.8) = .3932 + .2621 = .6553$.

b. Let $h(Y)$ = the number of available spaces. Then

when y is:	0	1	2	3	4	5	6
$h(y)$ is:	4	3	2	1	0	0	0

The expected number of available spaces when the limousine departs equals

$$E[h(Y)] = \sum_{y=0}^{6} h(y) \cdot b(y;6,.8) = 4(.000) + 3(.002) = 2(.015) + 1(.082) + 0 + 0 + 0 = 0.118.$$

c. Possible X values are 0, 1, 2, 3, and 4. $X = 0$ if there are 3 reservations and 0 show or 4 reservations and 0 show or 5 reservations and 0 show or 6 reservations and none show, so
$P(X = 0) = b(0; 3, .8)(.1) + b(0; 4, .8)(.2) + b(0; 5, .8)(.3) + b(0; 6, .8)(.4)$
$= .0080(.1) + .0016(.2) + .0003(.3) + .0001(.4) = .0013$.
Similarly, $P(X = 1) = b(1; 3, .8)(.1) + \ldots + b(1; 6, .8)(.4) = .0172$; $P(X = 2) = .0906$;
$P(X = 3) = .2273$; and $P(X = 4) = .6636$.

These values are displayed below.

x	0	1	2	3	4
$p(x)$	.0013	.0172	.0906	.2273	.6636

68.

a. There are 18 items (people) total, 8 of which are "successes" (first-time examinees). Among these 18 items, 6 have been randomly assigned to this particular examiner. So, the random variable X is hypergeometric, with $N = 18$, $M = 8$, and $n = 6$.

b. $$P(X = 2) = \frac{\binom{8}{2}\binom{18-8}{6-2}}{\binom{18}{6}} = \frac{\binom{8}{2}\binom{10}{4}}{\binom{18}{6}} = \frac{(28)(210)}{18564} = .3167.$$

$$P(X \le 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{\binom{8}{0}\binom{10}{6}}{\binom{18}{6}} + \frac{\binom{8}{1}\binom{10}{5}}{\binom{18}{6}} + .3167 =$$

$.0113 + .1086 + .3167 = .4366$.
$P(X \ge 2) = 1 - P(X \le 1) = 1 - [P(X = 0) + P(X = 1)] = 1 - [.0113 + .1086] = .8801$.

c. $E(X) = n \cdot \frac{M}{N} = 6 \cdot \frac{8}{18} = 2.67$; $V(X) = \left(\frac{18-6}{18-1}\right) \cdot 6\left(\frac{8}{18}\right)\left(1 - \frac{8}{18}\right) = 1.04575$; $\sigma = 1.023$.

72.

a. There are $N = 11$ candidates, $M = 4$ in the "top four" (obviously), and $n = 6$ selected for the first day's interviews. So, the probability x of the "top four" are interviewed on the first day equals $h(x; 6, 4, 11) =$

$$\frac{\binom{4}{x}\binom{7}{6-x}}{\binom{11}{6}}.$$

b. With X = the number of "top four" interview candidates on the first day, $E(X) = n \cdot \frac{M}{N} = 6 \cdot \frac{4}{11} = 2.18$.

76. This question relates to the negative binomial distribution, but we can't use it directly (X, as it's defined, doesn't fit the negative binomial description). Instead, let's reason it out.
Clearly, X can't be 0, 1, or 2.
$P(X = 3) = P(BBB \text{ or } GGG) = (.5)^3 + (.5)^3 = .25$.
$P(X = 4) = P(BBGB \text{ or } BGBB \text{ or } GBBB \text{ or } GGBG \text{ or } GBGG \text{ or } BGGG) = 6(.5)^4 = .375$.
$P(X = 5)$ look scary until we realize that 5 is the maximum possible value of X!
(If you have 5 kids, you must have at least 3 of one gender.) So, $P(X = 5) = 1 - .25 - .375 = .375$, and that completes the pmf of X.

x	3	4	5
$p(x)$	.25	.375	.375

80. Solutions are found using the cumulative Poisson table, $F(x; \mu) = F(x; 4)$.

a. $P(X \le 4) = F(4; 4) = .629$, while $P(X < 4) = P(X \le 3) = F(3; 4) = .434$.

b. $P(4 \le X \le 8) = F(8; 4) - F(3; 4) = .545$.

c. $P(X \ge 8) = 1 - P(X < 8) = 1 - P(X \le 7) = 1 - F(7; 4) = .051$.

d. For this Poisson model, $\mu = 4$ and so $\sigma = \sqrt{4} = 2$. The desired probability is $P(X \le \mu + \sigma) = P(X \le 4 + 2) = P(X \le 6) = F(6; 4) = .889$.

82.

a. $P(X = 1) = F(1; .2) - F(0; .2) = .982 - .819 = .163$.

b. $P(X \ge 2) = 1 - P(X \le 1) = 1 - F(1; .2) = 1 - .982 = .018$.

c. The probability a disk contains zero missing pulses is $P(X = 0) = \frac{e^{-2}(.2)^0}{0!} = .819$. Since the two disks are independent, $P(1^{\text{st}} \text{ doesn't} \cap 2^{\text{nd}} \text{ doesn't}) = P(1^{\text{st}} \text{ doesn't}) \cdot P(2^{\text{nd}} \text{ doesn't})$
$= (.819)(.819) = .671$.

88. Let X = the number of diodes on a board that fail. Then $X \sim \text{Bin}(n = 200, p = .01)$.

a. $E(X) = np = (200)(.01) = 2$; $V(X) = npq = (200)(.01)(.99) = 1.98$, so $\sigma = 1.407$.

b. X has approximately a Poisson distribution with $\mu = np = 2$, so $P(X \geq 4) = 1 - P(X \leq 3) = 1 - F(3; 2) = 1 - .857 = .143$.

c. For any one board, $P(\text{board works properly}) = P(\text{all diodes work}) = P(X = 0) = \dfrac{e^{-2}2^0}{0!} = .135$.

Let Y = the number among the five boards that work, a binomial rv with $n = 5$ and $p = .135$.

Then $P(Y \geq 4) = P(Y = 4) + P(Y = 5) = \binom{5}{4}(.135)^4(.865) + \binom{5}{5}(.135)^5(.865)^0 = .00148$.